

HSRP

Hot Standby Routing
Protocol (HSRP)

HSRP

Address Resolution Protocol (ARP)

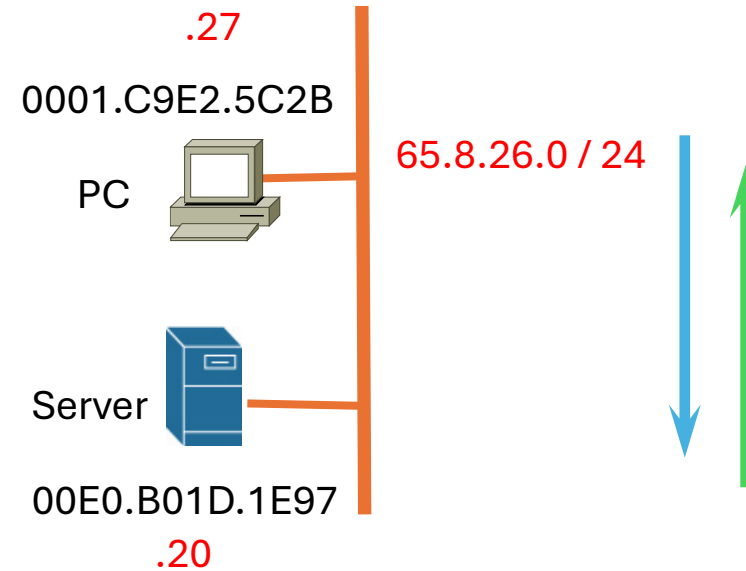
Address Resolution Protocol

Address Resolution Protocol (ARP)

When PC on a LAN wants to send a message to another system, it sends a request asking for the MAC address corresponding to the destination IP.

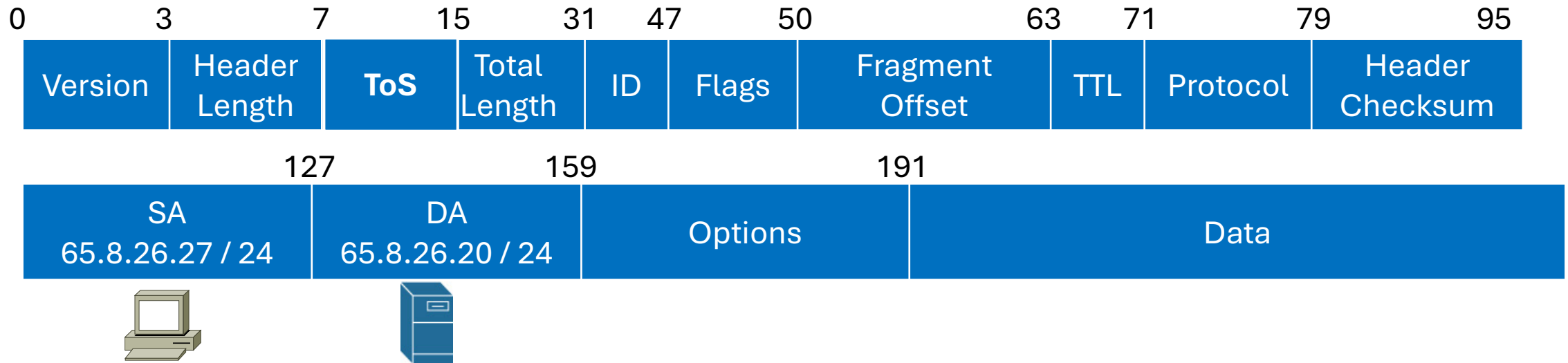
If the other system is on the same LAN it responds with its MAC address.

This process uses the **Address Resolution Protocol**.

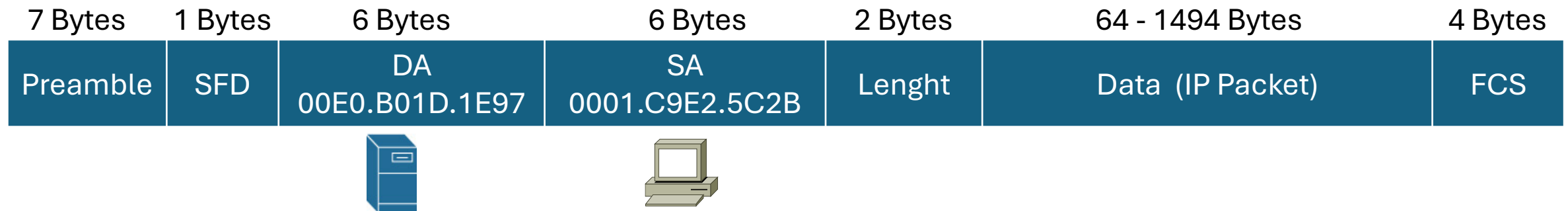


IPv4 Packet / Ethernet Frame

The IPv4 Packet Format



Ethernet Frame



HSRP Background

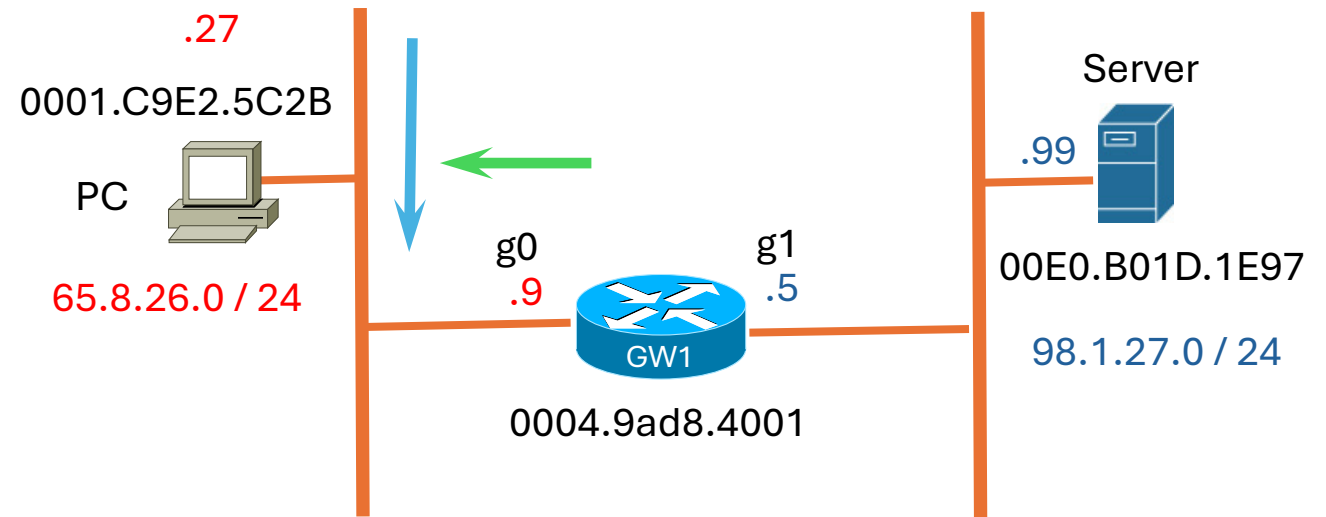
Proxy ARP

If the system belongs to another network

The PC still sends an ARP request.

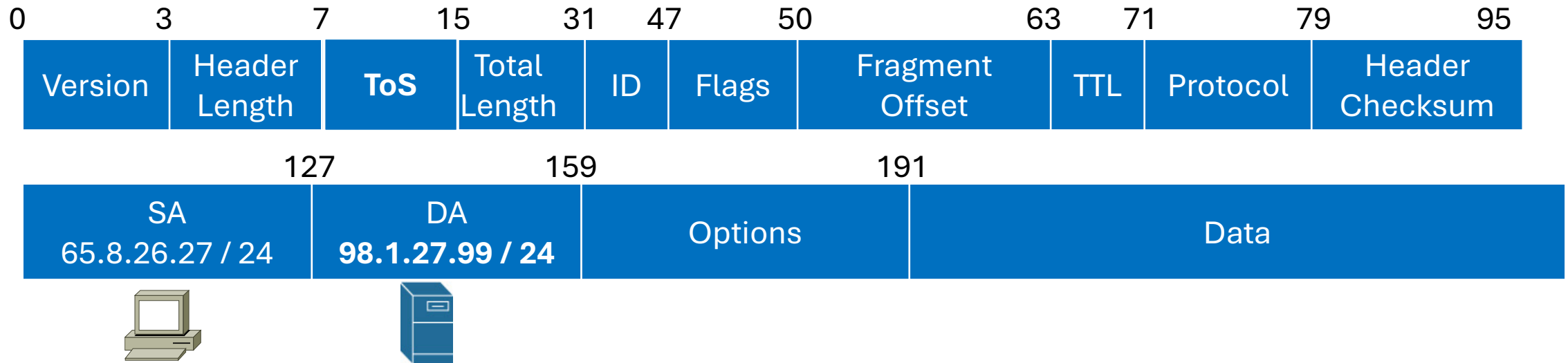
And the Router responds with its own MAC

This process is called **Proxy ARP**

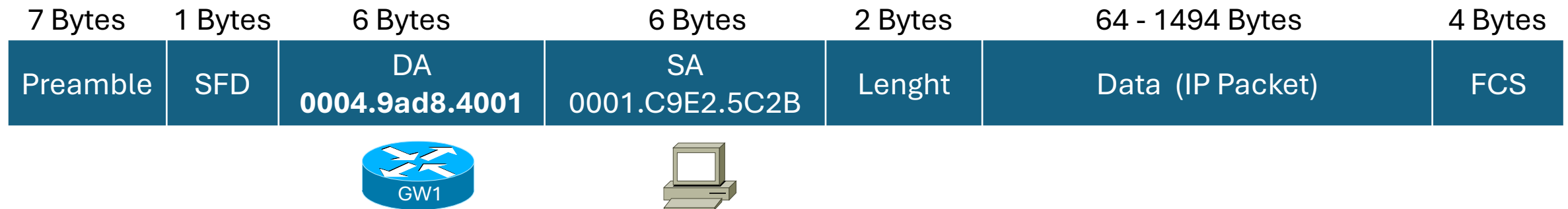


Proxy ARP

The IPv4 Packet Format



Ethernet Frame

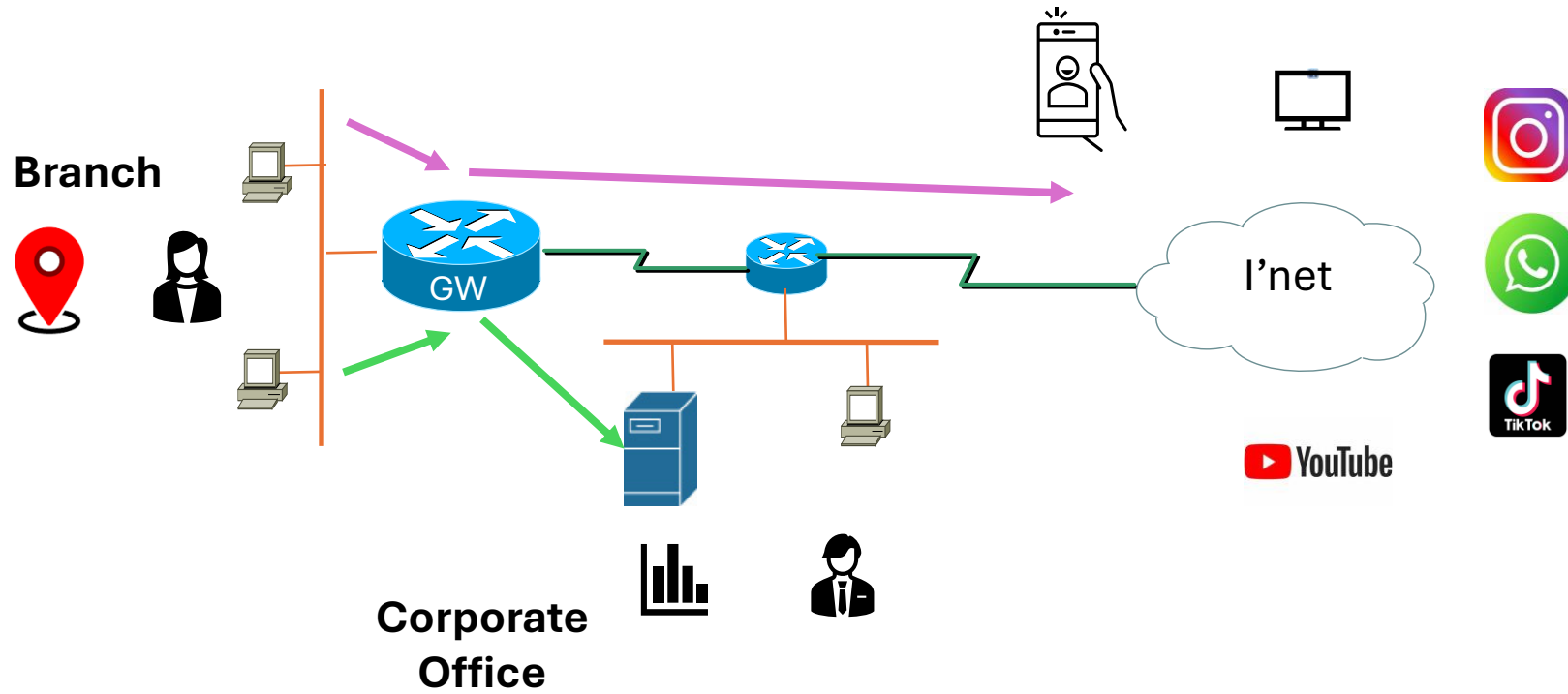


First Hop Redundancy Protocol

First Hop
Redundancy Protocol
(FHRP)

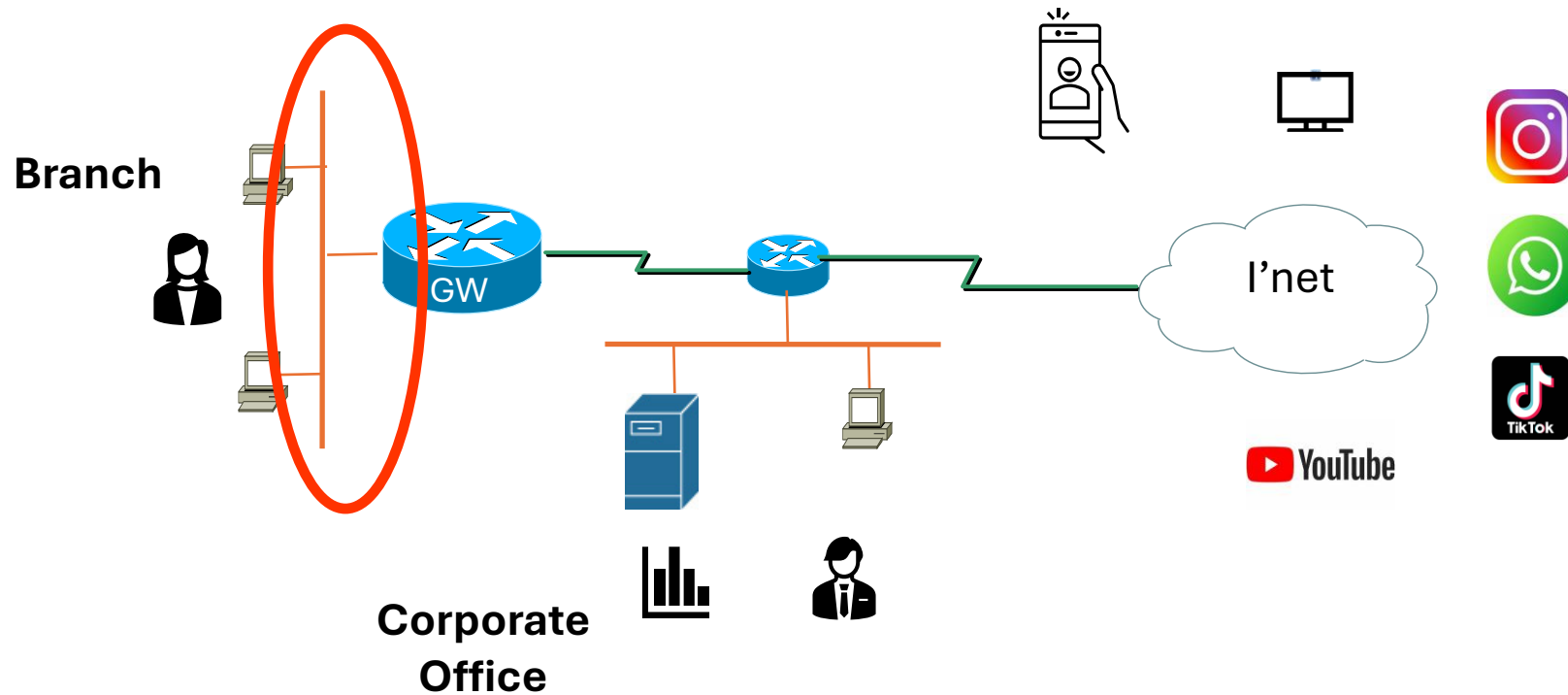
FHRP Background

In the IP world usually a user's device connected to a network is configured with a host IP address and a Gateway, which is a router, to be able to communicate with devices in other networks.



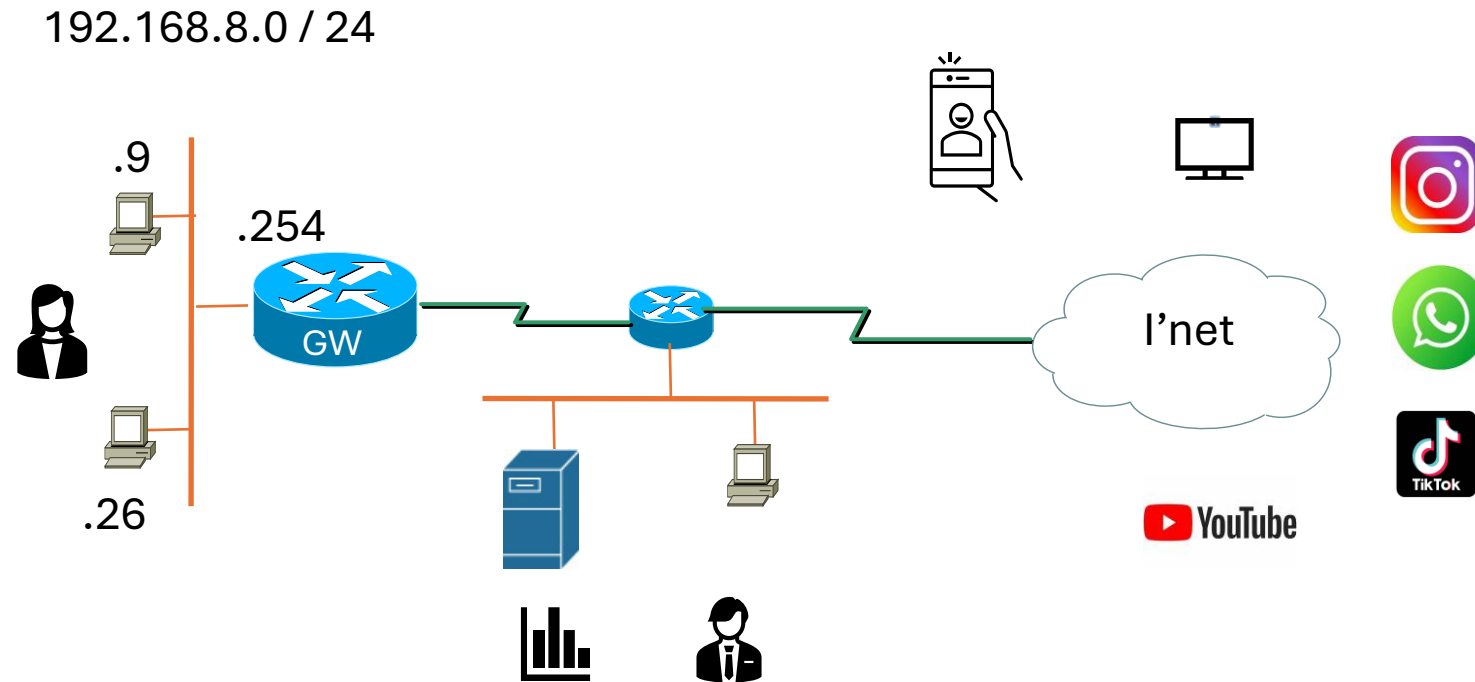
FHRP Background

The gateway's IP address must belong to the same IP network's address space of the host.



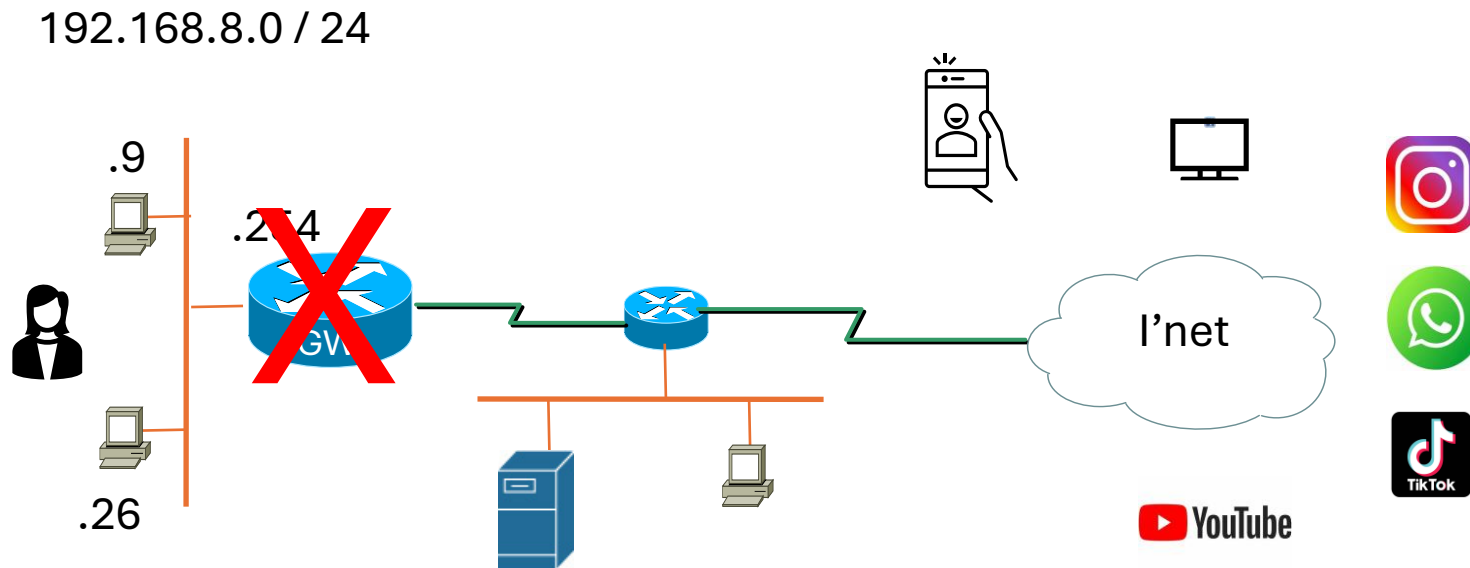
FHRP Background

The gateway address can be ANY valid host address within the network.



FHRP Background

What does it happen if the GW somehow goes down?

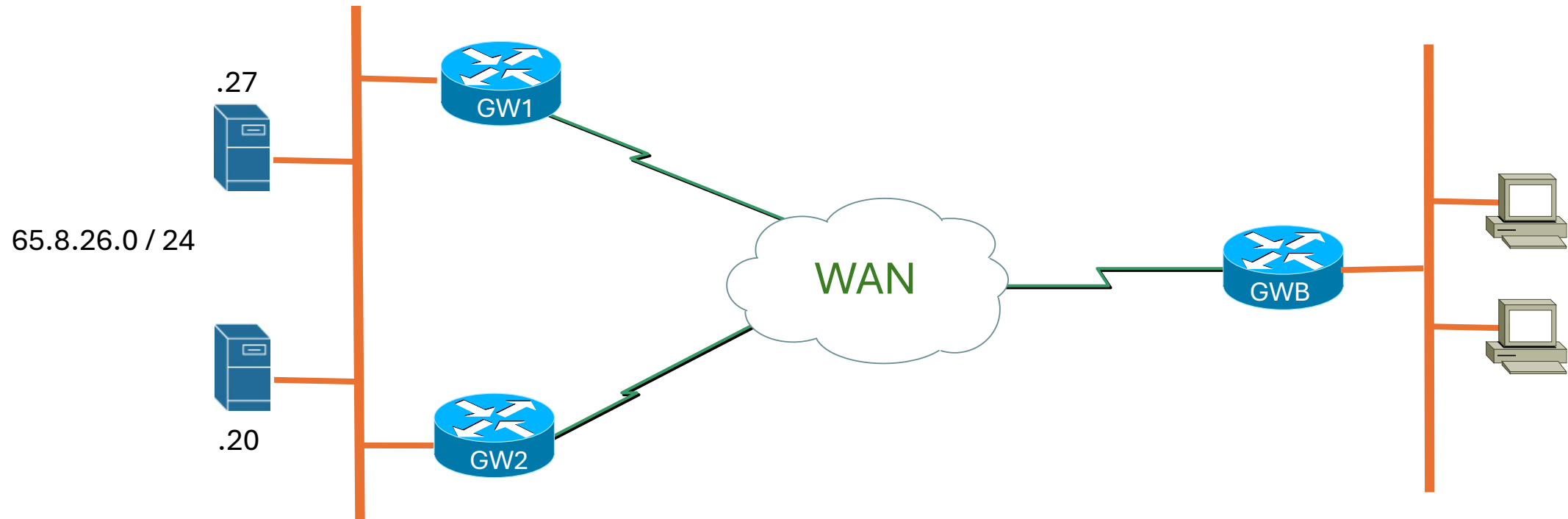


The ability a network has to recover from the failure of a device acting as a gateway is known as **First Hop Redundancy**.

FHRP Background

First Hop Redundancy Protocol (FHRP) ensures that user traffic immediately and transparently recovers from **First Hop** failures in network edge devices.

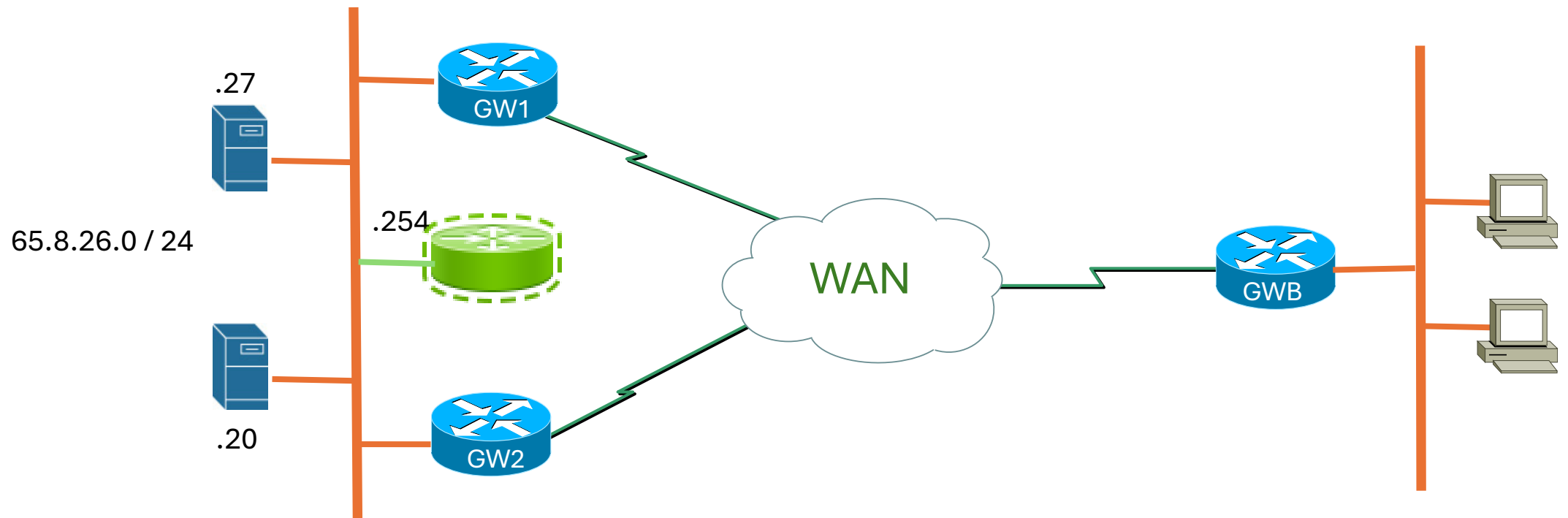
FHRP is a set of mechanisms created to provide redundancy in equipment and path for the gateway.



FHRP Background

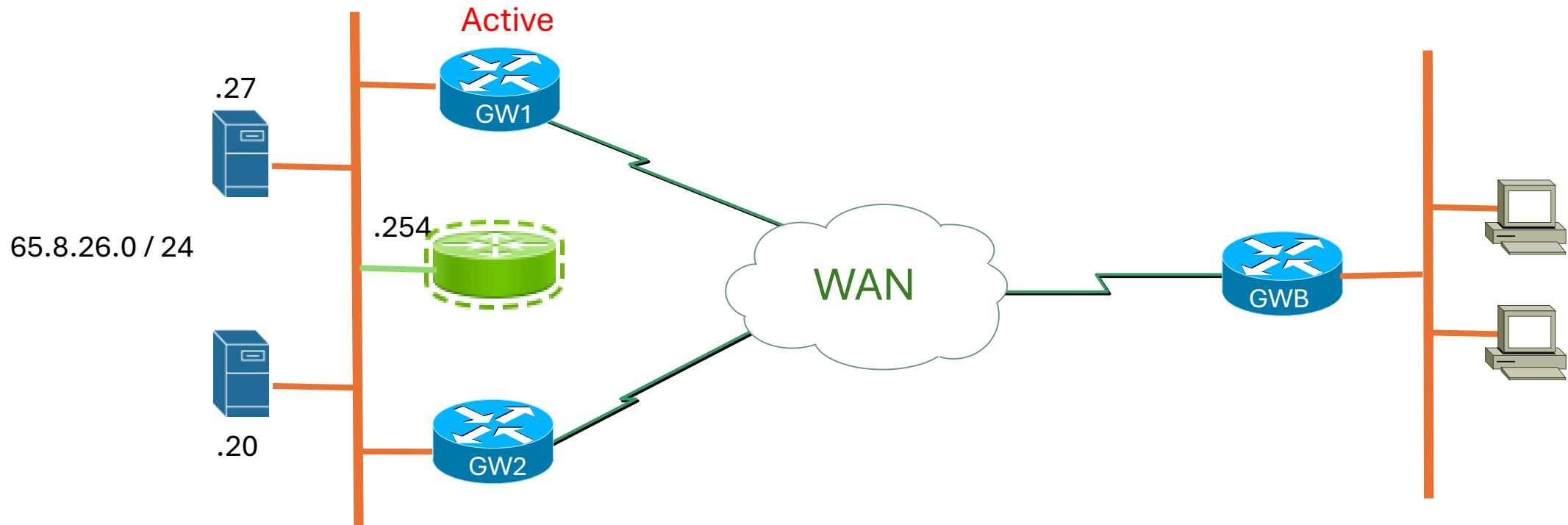
FHRP creates a virtual router that takes control of the gateway's IP address (virtual) and its virtual MAC within a group of routers.

This set of routers are known as FHRP group or Standby group.



FHRP Operation

A single router elected from the group is responsible for distribution of the packets that hosts send to the virtual router. This router is known as the **Active** router.

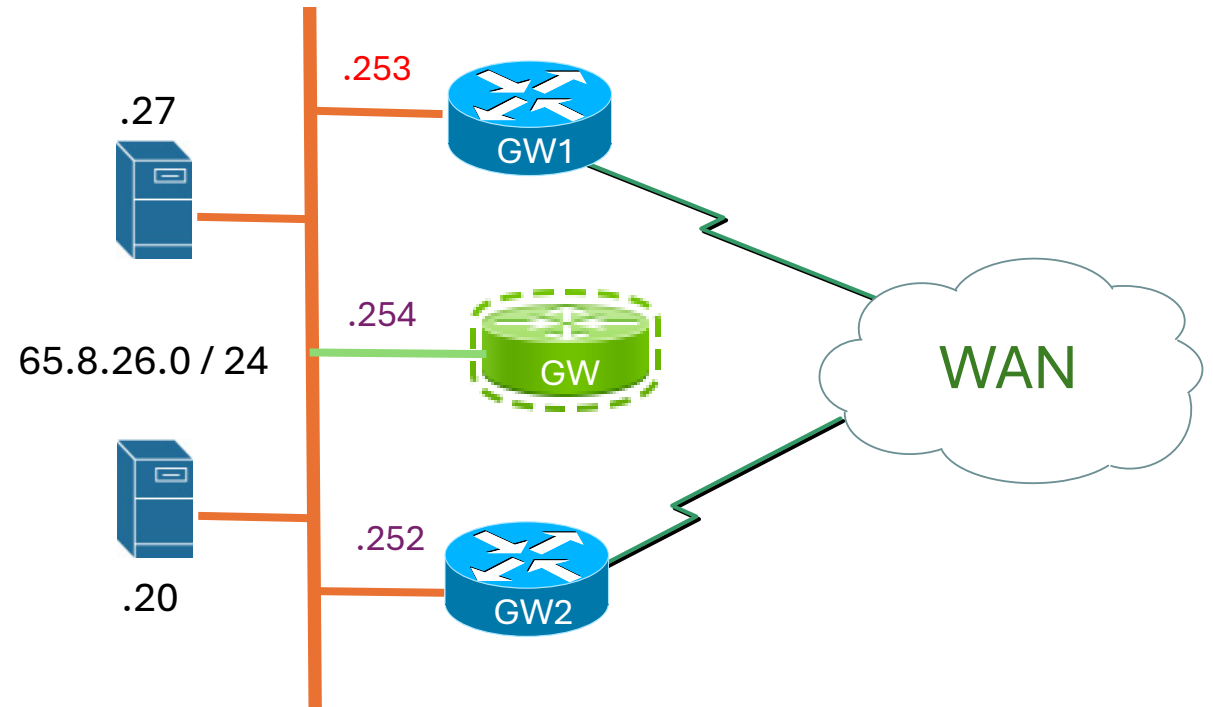


FHRP OPERATION

When an FHRP group is configured, there is an election process.

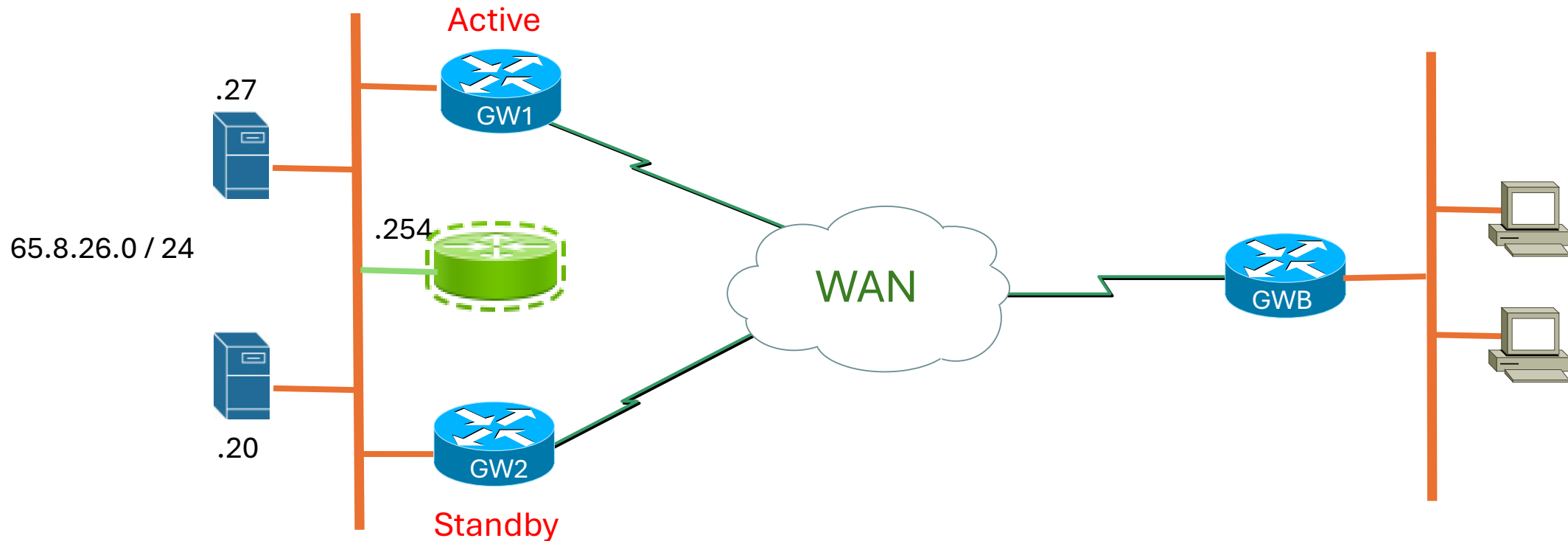
The router with the highest IP address becomes the Active Router.

The Active router election can be influenced by assigning a Priority to each router.



FHRP Operation

Another router is elected as the **Standby** router. In the event that the Active router fails, the Standby assumes the packet-forwarding duties of the Active router.

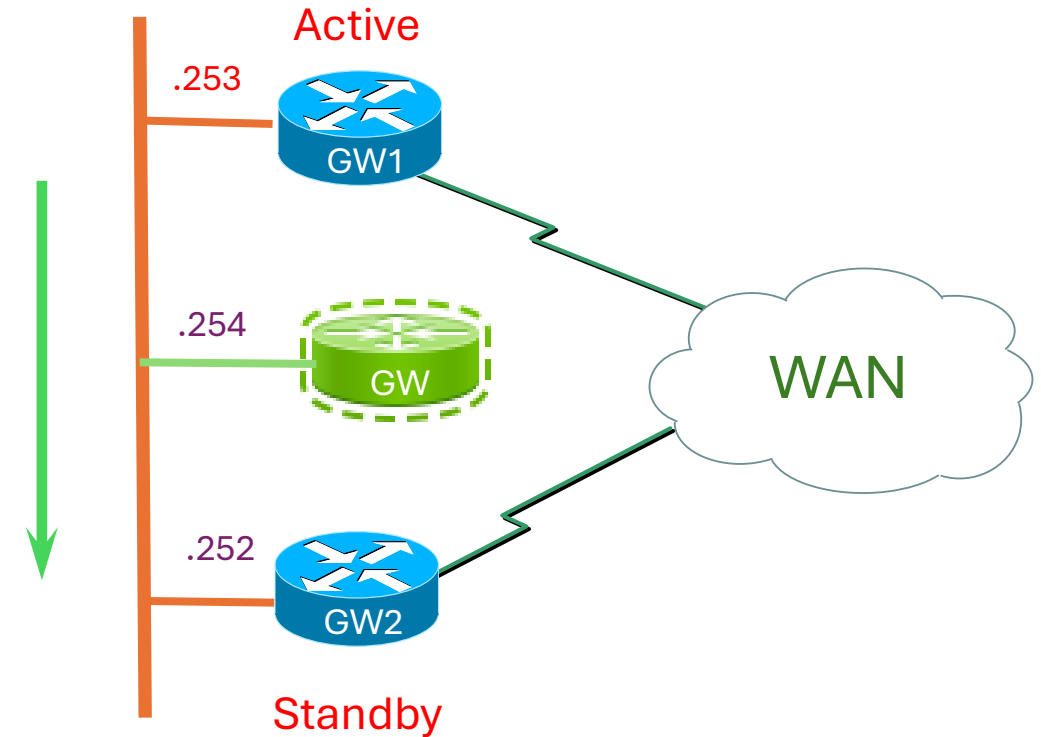


FHRP OPERATION

The Active and Standby routers exchange Hello packets to monitor each other.

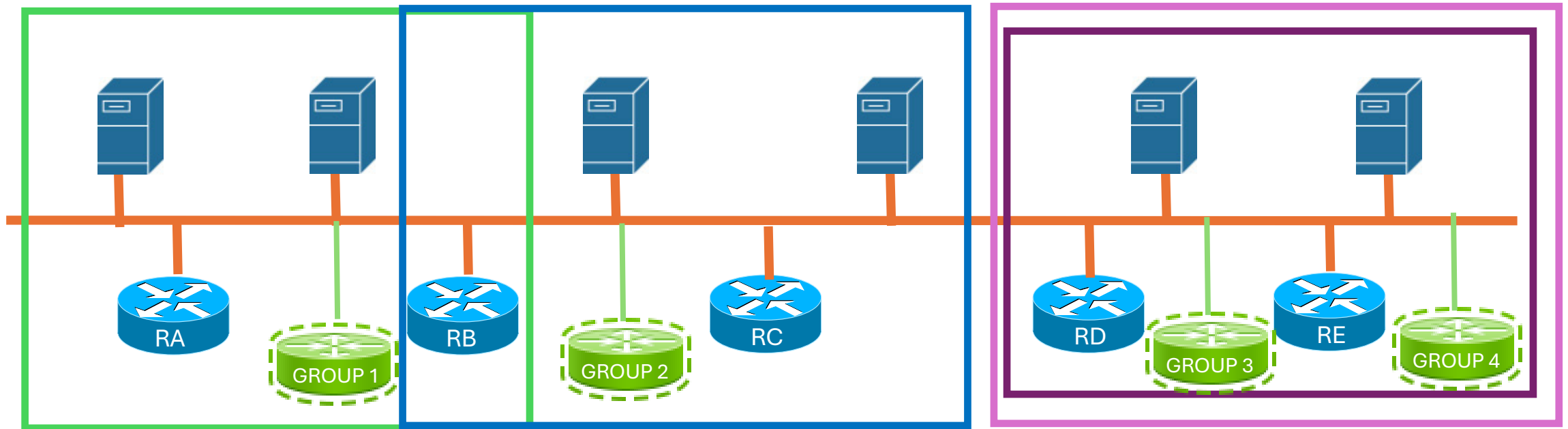
If the standby router stops receiving up to 3 hello packets, it assumes the active router is gone and takes control of the virtual IP and the virtual MAC addresses.

65.8.26.0 / 24



FHRP Operation

On a particular LAN, multiple FHRP groups can coexist and overlap. Each group emulates a single virtual router. The individual routers can participate in multiple groups.



HSRP

Hot Stand-By Routing Protocol (HSRP)

HSRP

Hot Standby Routing Protocol (HSRP) was the first protocol and it was developed by Cisco.

There are 2 versions of HSRP:

HSRP version 1 could handle up to 256 groups (it uses 8 bits).

HSRP version 2 could handle up to 4096 groups (it uses 12 bits).

HSRP version 1 used
0000.0c07.acNN (where NN is the HSRP group number)
As Virtual MAC address

HSRP Configuration

GW1

```
interface GigabitEthernet0/0/0  
ip address 10.10.10.10 255.255.255.0  
standby 0 ip 10.10.10.100
```

GW2

```
interface GigabitEthernet0/0/0  
ip address 10.10.10.20 255.255.255.0  
standby 0 ip 10.10.10.100
```

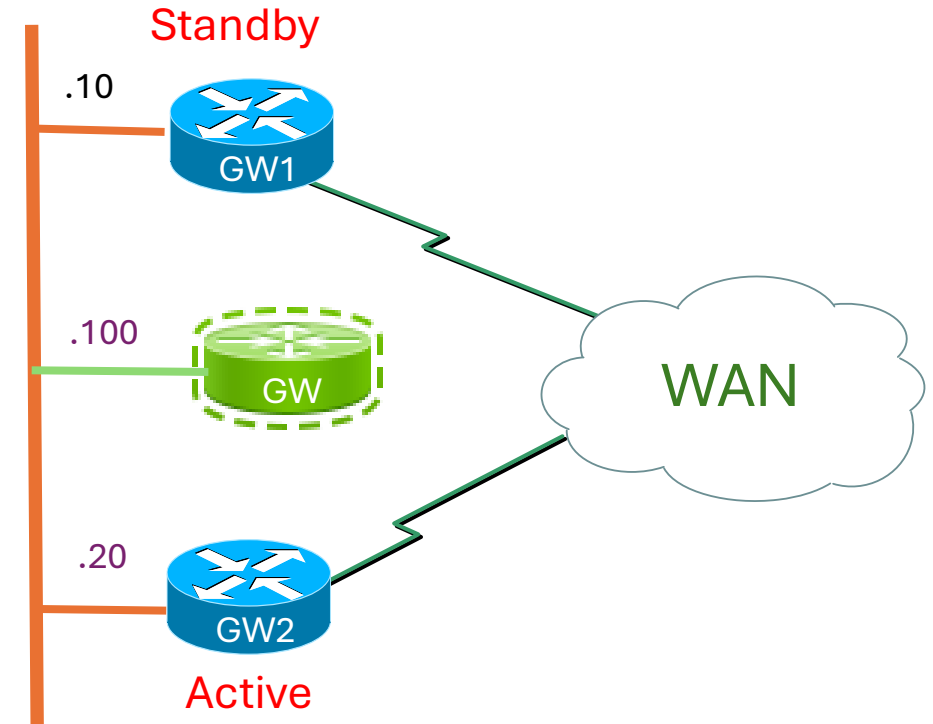
When configuring HSRP if:

No group is specified; the default is 0

No priority is specified; the default is 100

If both routers have the same priority, the active router will be the one with the highest IP address

10.10.10.0 / 24



HSRP Verification

Use the ***sh standby*** command to verify HSRP operation.

```
GW1#sh standby ?  
FastEthernet FastEthernet IEEE 802.3  
brief brief  
<cr>
```

```
GW1#show standby brief  
P indicates configured to preempt.  
|  
Interface Grp Pri P State Active Standby Virtual IP  
Gig0/0/0 0 100 Standby 10.10.10.20 local 10.10.10.100
```

HSRP Verification

Use the ***sh standby*** command to verify HSRP operation.

```
GW1#sh standby
GigabitEthernet0/0/0 - Group 0
State is Standby
13 state changes, last state change 00:27:11
Virtual IP address is 10.10.10.100
Active virtual MAC address is 0000.0C07.AC00
Local virtual MAC address is 0000.0C07.AC00 (v1 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 1.525 secs
Preemption disabled
Active router is 10.10.10.20
Standby router is local
Priority 100 (default 100)
Group name is hsrp-Gig0/0/0-0 (default)
```

HSRP Configuration

Use the **priority** command to select an active router.

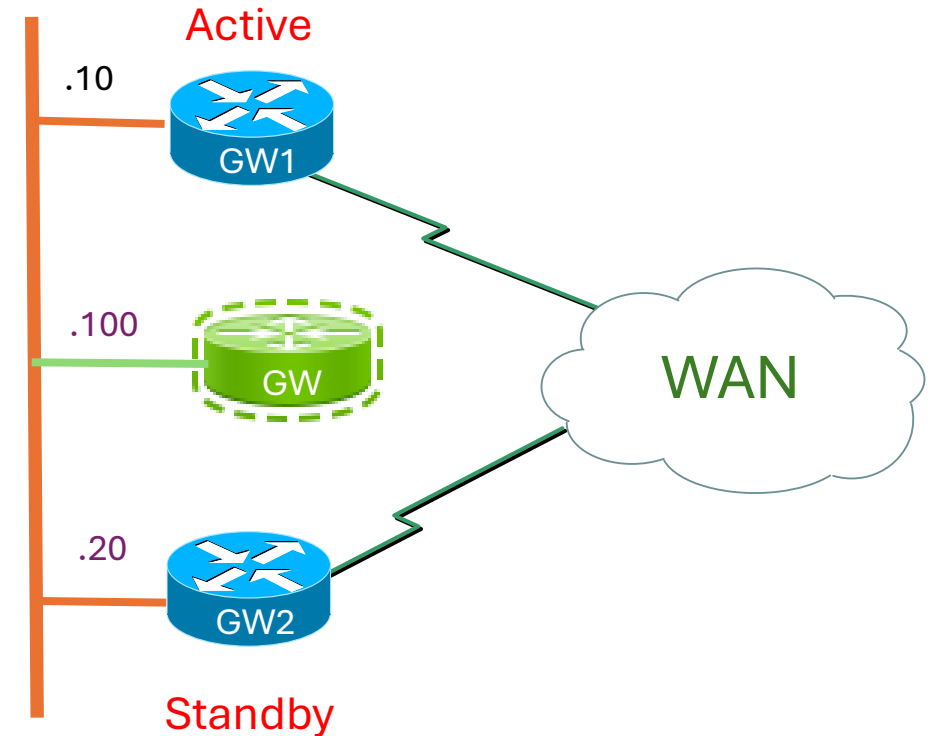
GW1

```
interface GigabitEthernet0/0/0
ip address 10.10.10.10 255.255.255.0
standby 0 ip 10.10.10.100
standby 0 priority 105
```

GW2

```
interface GigabitEthernet0/0/0
ip address 10.10.10.20 255.255.255.0
standby 0 ip 10.10.10.100
standby priority 100
```

10.10.10.0 / 24



HSRP Verification

Use the **sh standby** command to verify HSRP operation.

```
GW1#show standby brief
```

```
      P indicates configured to preempt.
```

```
|
```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Gig0/0/0	0	105		Active	local	10.10.10.20	10.10.10.100

```
GW2#show standby brief
```

```
      P indicates configured to preempt.
```

```
|
```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Gig0/0/0	0	100		Standby	10.10.10.10	local	10.10.10.100

HSRP Configuration

Use the ***preempt*** command to make the highest priority router override an active lower priority router.

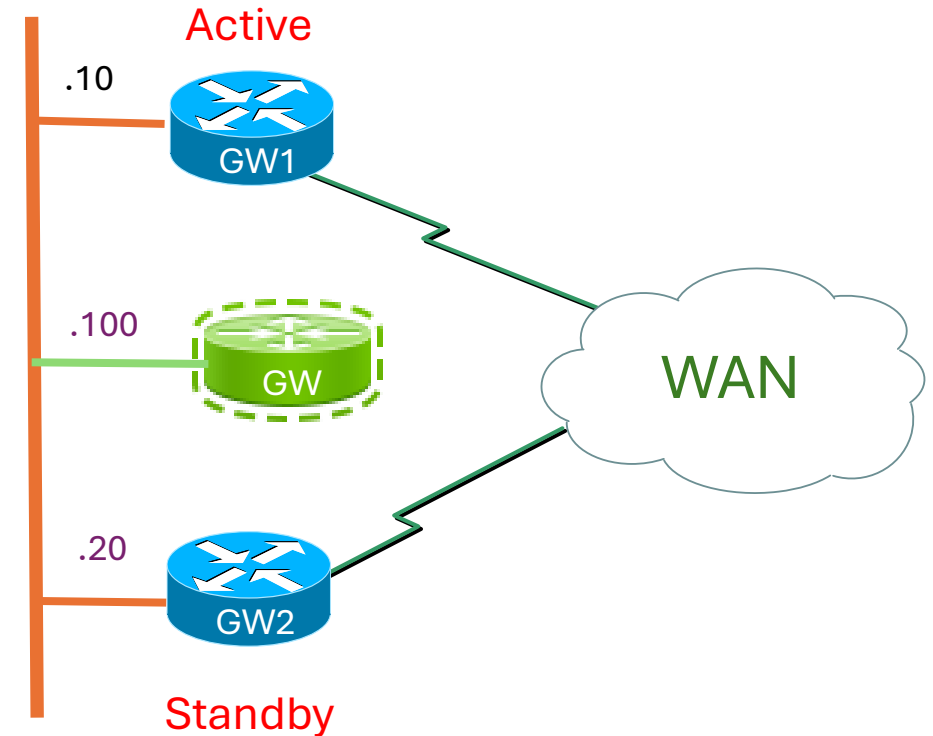
GW1

```
interface GigabitEthernet0/0/0
ip address 10.10.10.10 255.255.255.0
standby 0 ip 10.10.10.100
standby 0 priority 105
standby 0 preempt
```

GW2

```
interface GigabitEthernet0/0/0
ip address 10.10.10.20 255.255.255.0
standby 0 ip 10.10.10.100
standby 0 priority 100
```

10.10.10.0 / 24



HSRP Verification

Use the ***sh standby*** command to verify HSRP operation.

```
GW1(config-if)#int g0/0/0
```

```
GW1(config-if)#shut
```

```
GW1(config-if)#
```

```
%HSRP-6-STATECHANGE: GigabitEthernet0/0/0 Grp 0 state Active -> Init
```

```
GW1(config-if)#int g0/0/0
```

```
GW1(config-if)#no shut
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
```

```
%HSRP-6-STATECHANGE: GigabitEthernet0/0/0 Grp 0 state Speak -> Standby
```

```
%HSRP-6-STATECHANGE: GigabitEthernet0/0/0 Grp 0 state Standby -> Active
```

HSRP Configuration

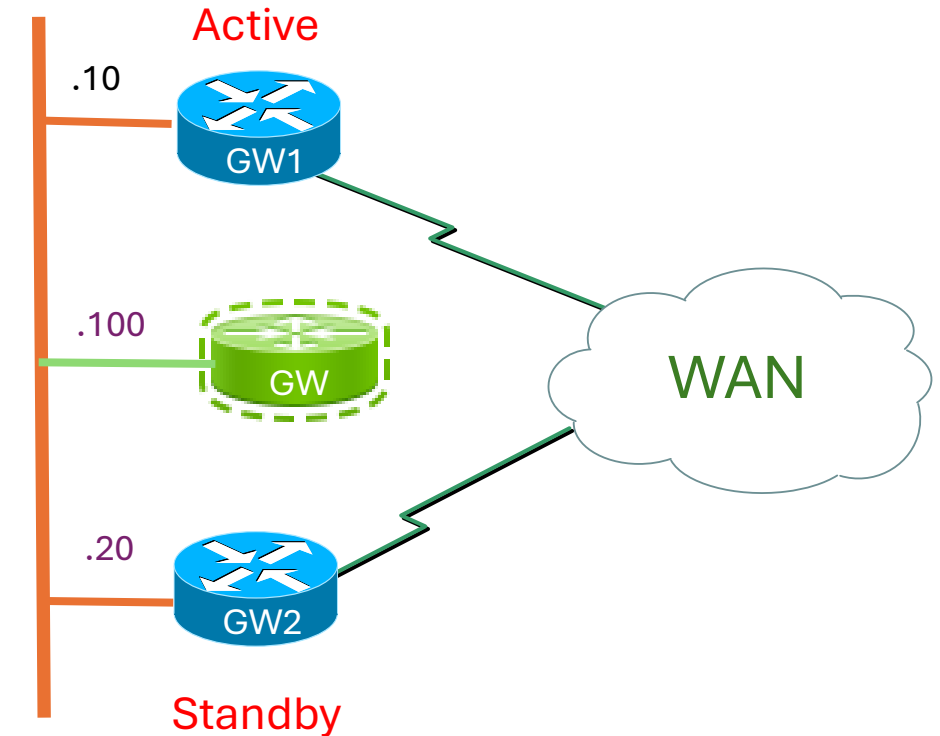
Use the ***track [interface]*** command to monitor such interface and allow the standby route to become active.

GW1

```
interface GigabitEthernet0/0/0
ip address 10.10.10.10 255.255.255.0
standby 0 ip 10.10.10.100
standby 0 priority 105
standby 0 preempt
standby 0 track Serial0/1/0
```

The ***track*** command decreases the interface priority by 10 units. If this amount causes the active router's priority to be lower than the standby router, the standby router will become active.

10.10.10.0 / 24



HSRP Configuration

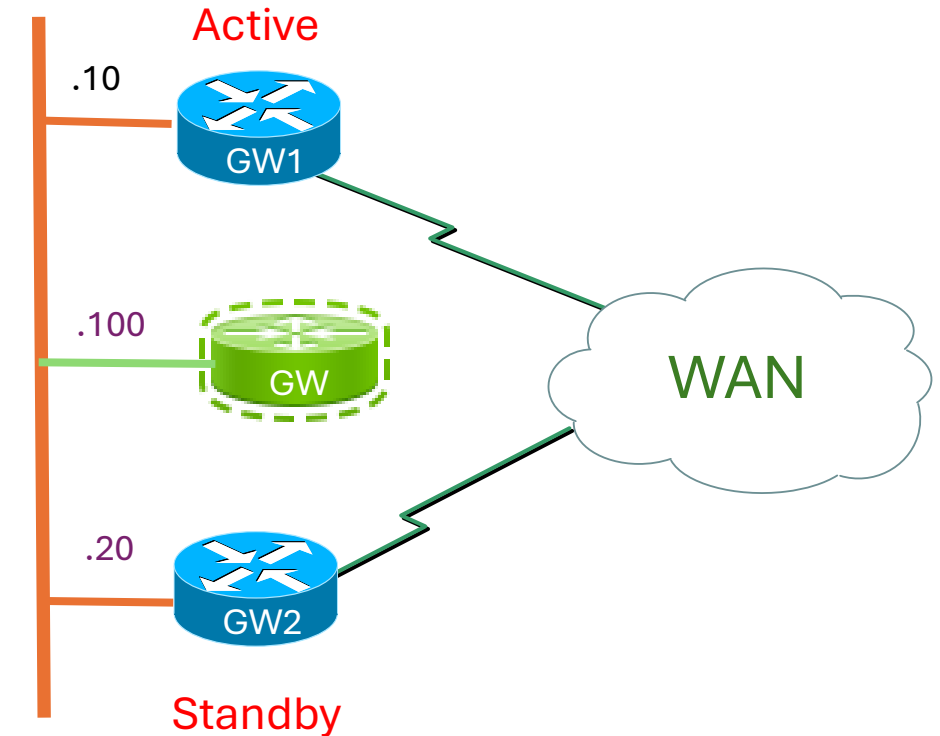
In some IOS versions this feature configuration changes.

A track object must be created in general configuration mode and then must be associated to the interface.

```
GW1(config)#track 10 int serial1/0 ip routing  
GW1(config-track)#ex
```

```
interface GigabitEthernet0/0/0  
ip address 10.10.10.10 255.255.255.0  
standby 0 ip 10.10.10.100  
standby 0 priority 105  
standby 0 preempt  
standby 0 track 10 decrement 10
```

10.10.10.0 / 24



HSRP Verification

Use the ***sh standby*** command to verify HSRP operation.

```
GW1(config-if)#int s0/1/0
GW1(config-if)#shut

%HSRP-6-STATECHANGE: GigabitEthernet0/0/0 Grp 0 state Speak -> Standby
```

First Hop Redundancy Protocol

First Hop
Redundancy Protocol
(FHRP)

First Hop Redundancy Protocol

There are other mechanisms in the FHRP suite:

FHRP Options	Description
HSRP IPv4	Hot Standby Routing Protocol. Uses an active router which routes packets of the LAN while a standby router waits and takes over in case the active router fails.
HSRP IPv6	HSRP for IPv6.
VRRP v2	Virtual Router Redundancy Protocol version 2. Similar to HSRP. The main difference is that a “Master Virtual Router” is dynamically assigned and there are several backup routers that can take over in case the master fails. VRRP is standard.
VRRP v3	VRRP for IPv6.
GLBP IPv4	Gateway Load Balancing Protocol is Cisco’s proprietary. Protects data Exchange as HSRP and VRRP. It adds the ability to use several active routers that share load and are backup for one another.
GLBP IPv6	GLBP for IPv6.
ICMP’s IRDP	ICMP’s Router Discovery Protocol allows to locate IPv4 Routers that provide connectivity to other non-local networks.

Thank You

Thank You